CHE 201 Spring 2020 Quiz 2

Thanh Nguyen

TOTAL POINTS

100 / 100

QUESTION 1

1 40 pts

1.1 1a **20 / 20**

✓ **- 0 pts** **Correct**

- 5 pts not explained

1.2 1b **20 / 20**

✓ **- 0 pts** **Correct**

- 5 pts v1 and v2 are incorrect

- 5 pts state 1 not shown

QUESTION 2

2 30 pts

2.1 2a **15 / 15**

✓ **- 0 pts** **Correct**

- 2 pts calculation error

2.2 2b **15 / 15**

✓ **- 0 pts** **Correct**

- 2 pts calculation error

QUESTION 3

3 3 **30 / 30**

✓ **- 0 pts** **Correct**

- 3 pts not explained

- 3 pts calculation error, Q is -2kj/s

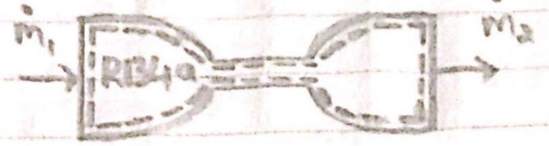
*CHE 205 - QUIZ 2

1.) Sketch

Inlet

. Saturated liquid

. $T_1 = 115°F$



Outlet

. $p_2 = 60 lbf/in^2$

Engineering Model

1. The system is a control volume at a steady state with 1 inlet and 1 outlet
2. $\Delta KE = \Delta PE = Q_{cv} = W_{cv} = 0$

Analysis

Mass balance $\frac{dm_{cv}}{dt} = \sum \dot{m}_i - \sum \dot{m}_e = 0 \rightarrow \dot{m}_1 - \dot{m}_2 = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

Energy balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \cancel{\dot{Q}_{cv}}^{0} - \cancel{\dot{W}_{cv}}^{0} + \dot{m}[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}}^{0} + \cancel{g(z_1 - z_2)}^{0}] = 0$

$\rightarrow \dot{m}(h_1 - h_2) = 0 \rightarrow h_1 = h_2$

a. Using table A-10E, saturated liquid at $T_1 = 115°F \rightarrow h_1 = 49.63$ Btu/lb,

$p_1 = 173.10$ lbf/in², $v_1 = 0.01429 ft^3/lb$

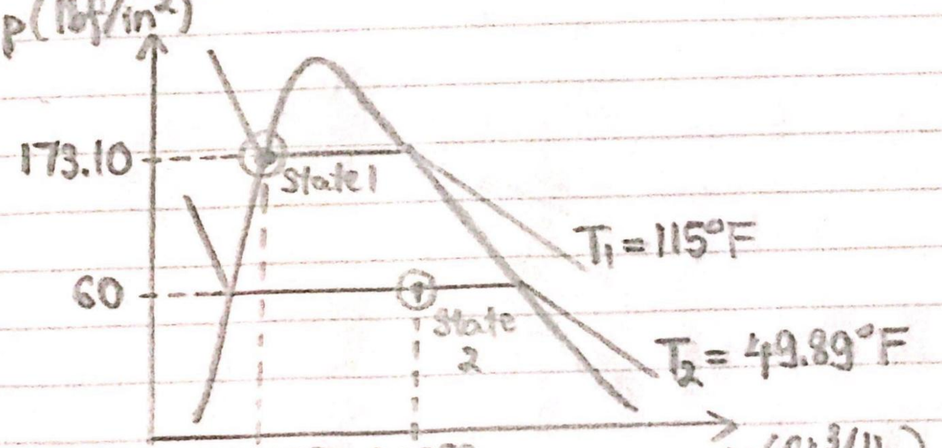
$\rightarrow h_2 = h_1 = 49.63$ Btu/lb

Using table A-11E, liquid-vapor mixture at $p_2 = 60 lbf/in^2 \rightarrow T_2 = 49.89°F$,

$h_f = 27.24$ Btu/lb, $h_g = 108.72$ Btu/lb, $v_{f2} = 0.01270 ft^3/lb$, $v_{g2} = 0.7887 ft^3/lb$

$\rightarrow h_2 = (1-x)h_f + xh_g \rightarrow x = \frac{h_2 - h_f}{h_g - h_f} = \frac{49.63 - 27.24}{108.72 - 27.24} = 0.2748$

b. $v_2 = (1-x)v_{f2} + x v_{g2} = (1-0.2748)(0.0127 ft^3/lb) + (0.2748)(0.7887 ft^3/lb)$
$= 0.2259 ft^3/lb$



1.1 1a 20 / 20

- ✓ - **0 pts** Correct
- - **5 pts** not explained

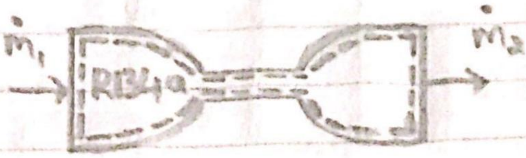
* CHE 205 - QUIZ 2

1.) <u>Sketch</u>

<u>Inlet</u>

. Saturated liquid

. $T_1 = 115°F$



Outlet

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1. The system is a control volume at a steady state with 1 inlet and 1 outlet
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$\rightarrow \dot{m}(h_1 - h_2) = 0 \rightarrow h_1 = h_2$

a. Using table A-10E, saturated liquid at $T_1 = 115°F \rightarrow h_1 = 49.63$ Btu/lb,

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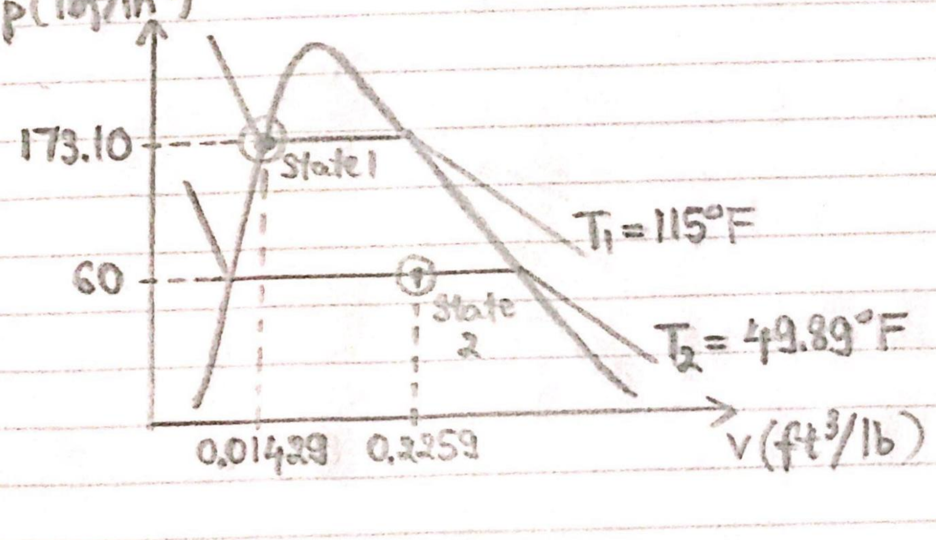

1.2 1b 20 / 20

- ✓ **- 0 pts** Correct
- **- 5 pts** v1 and v2 are incorrect
- **- 5 pts** state 1 not shown

2.) Sketch

$AV = 85L/min$
$T = 22°C$
$p = 3.6 \text{ bar}$

Engineering Model

1. The system is a control volume at a steady state with 1 inlet and 1 outlet
2. Ideal gas behavior
3. $\dot{W}_{cv} = \dot{Q}_{cv} = \Delta PE = 0$

Analysis

a. Volumetric flow rate $= 85L/min \times \frac{1m^3}{1000L} \times \frac{1min}{60s} = 1.4167\times10^{-3} m^3/s$

$$A = \frac{\pi D^2}{4} = \frac{\pi(0.08m)^2}{4} = 5.0265\times10^{-3} m^2$$

$$V = \frac{\text{Volumetric flow rate}}{A} = \frac{1.4167\times10^{-3} m^3/s}{5.0265\times10^{-3} m^2} = 0.2818 \text{ m/s}$$

b. $p\bar{v} = \bar{R}T \rightarrow pvM = \bar{R}T$

$$\rightarrow v = \frac{\bar{R}T}{pM} = \frac{(8.314kJ/Kmol.K)(22°C+273)}{(3.6bar)(28.97Kg/Kmol)} \times \frac{1bar}{10^2kPa} \times \frac{1KPa}{10^3 N/m^2} \times \frac{10^3 Nm}{1KJ}$$

$= 0.2352 m^3/Kg$

$$\dot{m} = \frac{AV}{v} = \frac{1.4167\times10^{-3} m^3/s}{0.2352 m^3/Kg} = 6.0234\times10^{-3} Kg/s$$

2.1 2a **15 / 15**

✓ **- 0 pts** Correct

- 2 pts calculation error

2.) Sketch

$AV = 85L/min$
$T = 22°C$
$p = 3.6\ bar$

Engineering Model
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b. $p\bar{v} = \bar{R}T \rightarrow pvM = \bar{R}T$

$$\rightarrow v = \frac{\bar{R}T}{pM} = \frac{(8.314kJ/kmol.K)(22°C + 273)}{(3.6bar)(28.97kg/kmol)} \times \frac{1bar}{10^2 kPa} \times \frac{1kPa}{10^3 N/m^2} \times \frac{10^3 Nm}{1kJ}$$

$= 0.2352\ m^3/kg$

$$\dot{m} = \frac{AV}{v} = \frac{1.4167 \times 10^{-3} m^3/s}{0.2352\ m^3/kg} = 6.0234 \times 10^{-3}\ kg/s$$

2.2 2b **15 / 15**

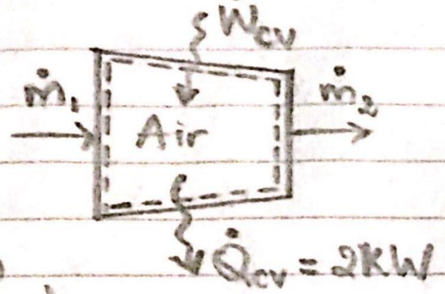
✓ **- 0 pts** Correct

- 2 pts calculation error

3.) Sketch

Inlet

- $p_1 = 1.05 \text{bar}$
- $T_1 = 300K$
- $A_1 V_1 = 12 m^3/min$



Outlet

- $p_2 = 12 \text{bar}$
- $T_2 = 400K$

Engineering Model

1. The system is a control volume at a steady state with 1 inlet and 1 outlet
2. Ideal gas behavior
3. $\Delta KE = \Delta PE = 0$

Analysis

Mass balance: $\frac{dm_{cv}}{dt} = \Sigma \dot{m}_i - \Sigma \dot{m}_e = 0 \rightarrow \dot{m}_1 = \dot{m}_2 = \dot{m}$

Energy balance: $\frac{dE_{cv}}{dt} = 0 \rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}\left[(h_1 - h_2) + \cancel{\frac{V_1^2 - V_2^2}{2}}^{\,0} + \cancel{g(z_1 - z_2)}^{\,0}\right] = 0$

$\rightarrow \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}(h_1 - h_2) = 0$

$p_1 \bar{v}_1 = \bar{R} T_1 \rightarrow p_1 v_1 M = \bar{R} T_1$

$\rightarrow v_1 = \frac{\bar{R} T_1}{p_1 M} = \frac{(8.314 \text{ kJ/kmol.K})(300K)}{(1.05 \text{bar})(28.97 \text{kg/kmol})} \times \frac{1 \text{bar}}{10^2 \text{KPa}} \times \frac{1 \text{KPa}}{10^3 N/m^2} \times \frac{10^3 Nm}{1 KJ}$

$= 0.820 \ m^3/Kg$

$\dot{m} = \frac{A_1 V_1}{v_1} = \frac{12 m^3/min}{0.820 m^3/kg} \times \frac{1 min}{60 s} = 0.2439 \text{ Kg/s}$

Using table A-22

- $T_1 = 300K \rightarrow h_1 = 300.19 \text{ KJ/Kg}$
- $T_2 = 400K \rightarrow h_2 = 400.98 \text{ KJ/Kg}$

$\rightarrow \dot{W}_{cv} = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$

$= -2KW + (0.2439 \text{ Kg/s})(300.19 \text{ KJ/kg} - 400.98 \text{ KJ/kg})$

$= -26.583$ KW on the system

3 3 **30 / 30**

✓ **- 0 pts** Correct

- 3 pts not explained

- 3 pts calculation error, Q is -2kj/s